

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks Available |     |     |       |       |      |
|----------|-----|------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                  | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 6.       | (a) | (i)  |  | Reduction in units = $15\,000 - 3\,000 = 12\,000$ [kWh] (1)<br>Reduction in cost = $12\,000 \text{ ecf} \times 20 = 240\,000$ [p] (1)<br>Conversion to £ = [£]2 400 (1)<br><br><b>Alternative:</b><br>$15\,000 \times 20 = 300\,000$ [p] (1)<br>$3\,000 \times 20 = 60\,000$ [p] (1)<br>Saving = $3\,000 \text{ ecf} - 600 \text{ ecf} = [£]2\,400$ (1)<br><br>Award 2 marks for answers of $2.4 \times 10^n$ where $n$ is not 3 |                 | 3   |     | 3     | 3     |      |
|          |     | (ii) |  | Payback time = $\frac{152\,400}{2\,400 \text{ ecf}} = 63.5$ [months] accept 64 [months]                                                                                                                                                                                                                                                                                                                                          |                 | 1   |     | 1     | 1     |      |
|          | (b) | (i)  |  | Selection and substitution into efficiency equation:<br>$\frac{1.9}{5.3}(1) [\times 100]$ (1)<br>% efficiency = 35.849 <b>or</b> 35.8 <b>or</b> 35.85 <b>or</b> 36 <b>or</b> 40 (1)<br>Award 1 mark for 0.36 or 0.4                                                                                                                                                                                                              |                 | 2   |     | 2     | 2     |      |
|          |     | (ii) |  | Weekly mass of dung = $150 \times 60 = 9\,000$ [kg/week] (1)<br>Weekly volume of gas = $9\,000 \text{ (ecf)} \times 0.05 = 450$ [m <sup>3</sup> ] (1)<br>Weekly number of kWh = $450 \text{ (ecf)} \times 1.9 = 855$ [kWh] (1)<br>Annual number of kWh = $855 \text{ (ecf)} \times 52$ (1)<br>= 44 460 <b>(ecf)</b> [kWh] [so farmer is incorrect] (1)                                                                           |                 |     | 5   | 5     | 5     |      |

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|----------|--|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |  |  |  | <p><b>Alternative 1</b><br/>Weekly mass of cow dung = 9 000 [kg] (1)<br/>Yearly mass of cow dung = 468 000 (<b>ecf</b>) (1)<br/>Gas/year = 468 000 (<b>ecf</b>) × 0.05 = 23 400 (1)<br/>Units/year 23 400 (<b>ecf</b>) × 1.9 (1)<br/>= 44 460 (<b>ecf</b>) [kWh] (1) [so incorrect]</p> <p><b>Alternative 2</b><br/>Weekly gas/cow = 60 × 0.05 = 3 (1)<br/>Weekly units/cow = 3 (<b>ecf</b>) × 1.9 = 5.7 (1)<br/>Weekly units for herd of cows = 5.7(<b>ecf</b>) × 150 = 855 (1)<br/>Units/year = 855 (<b>ecf</b>) × 52 (1)<br/>= 44 460 (<b>ecf</b>) [kWh] (1) [so incorrect]</p> <p><b>Alternative 3</b><br/>Weekly units required = <math>\frac{144\,000}{52} = 2\,769.23</math> (1)<br/>Weekly volume of gas required = <math>\frac{2\,769.23\text{ ecf}}{1.9} = 1\,457.5</math> (1)<br/>Weekly mass of dung required = <math>\frac{1\,457.5\text{ ecf}}{0.05} = 29\,149.8</math> (1)<br/>Number of cows required = <math>\frac{29\,149.8\text{ ecf}}{60}</math> (1)<br/>= 486 (<b>ecf</b>) (1) [so incorrect]</p> <p>Accept for 4 marks a final answer of 148 200 [kWh]<br/>Common answer 235 638 award 4 marks or 296.4 award 4 marks<br/>also or 124 020</p> |                 |     |     |       |       |      |

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|----------|-----|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|-----------|----------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                      | AO1             | AO2      | AO3      | Total     | Maths     | Prac     |
|          | (c) |  |  | <b>Any 2 × (1) from:</b><br>It reduces the amount of <u>methane</u> added to the atmosphere (1)<br><u>Methane</u> [into the atmosphere] is replaced with carbon dioxide or less harmful gas (1)<br><u>Methane</u> has a bigger impact on the greenhouse effect (1)<br>so <u>better</u> to use <u>methane</u> as a fuel <b>or</b> <u>better</u> to burn <u>methane</u> (1)<br>using the methane (or biogas) reduces demand for other fossil fuels (1) | 2               |          |          | 2         |           |          |
|          |     |  |  | <b>Question 6 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>2</b>        | <b>6</b> | <b>5</b> | <b>13</b> | <b>11</b> | <b>0</b> |